

Samuel Chau

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Summary

Computer Science graduate and maintainer of Profilarr, an open-source configuration-as-code tool for self-hosted media servers with 3.5M+ downloads and 2,400 stars. Experienced across full-stack web development (SvelteKit, Deno, TypeScript), systems programming, and cybersecurity, with a track record of designing developer tools that scale.

Education

University of Adelaide

Adelaide, SA

Bachelor in Computer Science, Major in Cybersecurity

Feb 2022 – Dec 2025

- GPA: **6.4/7.0** — Achieved 13 High Distinctions
- Top Results: Databases & Ethical Data (96), Web & Database Computing (95), Algorithm & Data Structure Analysis (93), Software Engineering & Project (92)

Experience

Maintainer & Lead Developer

Apr 2023 – present

Profilarr ([GitHub](#), [Website](#))

Open-source tool that brings configuration-as-code to self-hosted media servers.

- Designed a **CRDT-inspired, event-sourced configuration system** — config is stored as operations rather than final state, so upstream curators and local users can edit independently and merge without conflict; value guards catch same-field edits before they silently overwrite.
- Architected and built a **100K+ line SvelteKit/Deno/TypeScript/SQLite app solo**, including a Terraform-style sync engine (idempotent, retry-safe, with drift detection), a crash-recoverable job queue, and a fully type-safe data layer — TypeScript interfaces generated from the SQL schema and queried through Kysely, so schema, types, and queries can't fall out of sync.
- Built a reusable **component library** that leverages semantic CSS tokens to create a consistently styled, themable, and polished design, enabling rapid UI development. Enforced using custom Svelte-AST lint rules that ban raw HTML and hardcoded styles.
- Designed a **defense-in-depth security model that keeps raw secrets off the web surface entirely** — authenticated or not: clients receive presence flags rather than values, the app's own API key is hashed like a password (shown once, never retrievable), and backup downloads are credential-stripped before reaching the browser, leaving the filesystem as the only trust boundary for full-fidelity data. Layered with independent OWASP Top 10 protections: sanitized markdown, path-traversal and reverse-proxy-aware CSRF handling, and persistent rate limiting.
- Built a **CI/CD release pipeline** running build, lint, type checks, unit + integration tests against the production binary, Semgrep SAST, and Playwright E2E auth flows on every change.
- Grew to **3.5M+ downloads and 2,400 GitHub stars** ran all support, beta coordination, and contribution guidelines for a 2,000+ member community.

Projects

Synopsis (GitHub)

React, Express, LLMs, FFmpeg

AI Research Paper Video Platform

- Built full-stack platform using Grobid for paper parsing, LLMs for summarization/image generation, and FFmpeg for video synthesis.
- Pitched at **Tech eChallenge** startup competition; ranked **Top 5** out of 50+ teams.

OMesh (GitHub)*Python, Docker, WebSockets*

Secure Messaging Platform

- Built end-to-end encrypted messaging platform using low-level socket programming and custom protocol design.
- Implemented RSA/AES-GCM cryptographic pipeline for secure key exchange and message encryption.
- Deployed multi-server architecture with Docker Compose; real-time communication via WebSockets.

Rogue (GitHub)*C++, SFML*

Procedural Roguelike Game

- Built turn-based roguelike with procedural map generation and entity component system architecture.
- Implemented design patterns (Factory, Observer, State) for extensible game object behavior.

Skills

Languages: TypeScript, Python, C++, C#, SQL, HTML/CSS**Technologies:** SvelteKit, Deno, Docker, Git, Linux, SQLite, Kysely, Discord.js, FFmpeg**Practices:** GitOps, Event Sourcing, API Design, Open Source Development**Interests:** Self-Hosting, Homelabbing, Media Archival, Basketball, Technical Writing